

Automated Detection of Venous Beading in Fundus Images Using Patch-Based Deep Learning for Diabetic Retinopathy Staging

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Abstract—Venous beading is a clinically important biomarker of severe diabetic retinopathy but remains difficult to detect due to its subtle and localized vascular appearance. This paper presents an anatomically guided patch-based deep learning framework for venous beading detection in retinal fundus images. By integrating vessel masks with venous beading annotations, the proposed method extracts balanced 75×75 vessel-centered patches for CNN classification. Eleven CNN architectures are evaluated. EfficientNet-B0 achieves the highest accuracy on the main patch dataset, while ResNet-50 demonstrates the strongest balanced performance in a limited reserved-image evaluation. These results suggest that localized patch-level learning is effective for lesion-level venous beading detection and may support future interpretable diabetic retinopathy assessment systems.

Keywords—Venous Beading, Diabetic Retinopathy, Deep Learning, Convolutional Neural Networks, Retinal Image Analysis

I. INTRODUCTION

Diabetic retinopathy (DR) is a leading cause of preventable blindness, making early detection increasingly important worldwide. As DR progresses, retinal microvascular changes appear, including microaneurysms, hemorrhages, IRMA, and venous beading (VB). Among them, VB is a key biomarker of severe non-proliferative DR (NPDR) and progression toward proliferative DR (PDR) [1]. Despite its clinical importance, VB is difficult to identify consistently because it appears as subtle, irregular bead-like dilation and narrowing along retinal veins. As shown in Fig. 1, the vessel segments between the white arrow markers illustrate examples of venous beading. Its appearance varies across patients, disease stages, and imaging conditions, contributing to inter-rater variability and inconsistent grading [1], [2]. Although existing deep learning systems have achieved strong performance for DR detection, most focus on global image classification and do not explicitly localize specific lesions [3], [4]. Lesion-level datasets such as IDRiD highlight the need for localized abnormality detection [5]. As a result, subtle localized abnormalities such as VB may be overlooked.

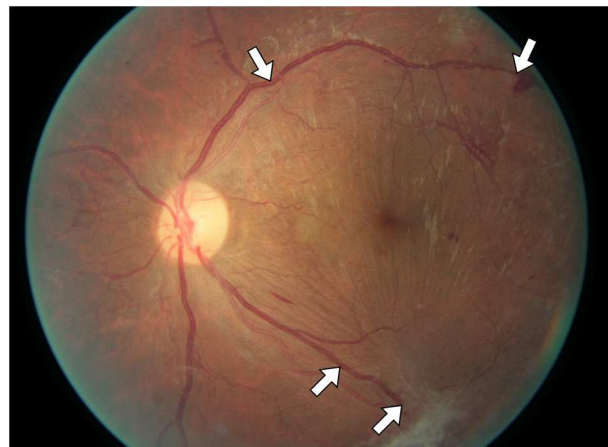


Fig. 1. Example of venous beading in a retinal fundus image.

This paper presents a patch-based deep learning framework for automated VB detection. It integrates blood vessel (BV) ground-truth masks, VB annotations, and intersection-based localization to isolate VB-specific vessel regions. Balanced 75×75 vessel-centered patches are extracted for CNN classification. The main contributions are: (1) anatomically valid VB patch extraction through mask integration; (2) balanced patch construction with deterministic augmentation; (3) evaluation of eleven CNN architectures on the main patch dataset; and (4) limited reserved dataset evaluation showing that ResNet-50 provides the strongest balanced performance on held-out images.

II. METHOD

A. Overview

The proposed framework adopts a patch-based learning strategy for venous beading (VB) detection. Instead of performing full-image classification, the method focuses on localized vessel regions where VB occurs. As illustrated in Fig. 2, the workflow consists of three stages: (1) data preparation, (2) model training/testing, and (3) reserved dataset evaluation. The

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main patch dataset is used to train and compare eleven CNN architectures, while the reserved dataset is kept separate to provide an additional held-out assessment of the top-performing models on unseen cases.

B. Dataset Description and Preparation

The dataset consists of retinal fundus images corresponding to Stage IV diabetic retinopathy, based on the International Clinical Diabetic Retinopathy and Diabetic Macular Edema Disease Severity Scales [6]. In this grading system, venous beading (VB) is one of the important diagnostic indicators of disease severity, although not every Stage IV image necessarily contains visible or usable VB regions. The initial dataset contains 31 Stage IV images. The VB annotations are manually labeled by an ophthalmologist, while blood vessel (BV) ground-truth masks are manually labeled by a domain expert for prior BV detection evaluation [7]. Vessel masks provide useful anatomical constraints for vessel-centered lesion localization in retinal image analysis [8]. However, not all 31 images are suitable for this study. Some Stage IV images do not contain usable VB regions for positive patch extraction, and a small number of images do not have complete BV ground-truth masks. These images are excluded because the proposed method requires both valid VB annotations and complete BV masks for vessel-centered patch construction. After filtering, 24 images remain for this study: 22 images are used to construct the main patch dataset for training and testing, and 2 images are reserved as an independent held-out dataset for limited reserved evaluation.

C. Venous Beading Region Localization

The BV ground-truth mask and the VB annotation mask are labeled separately. Although VB occurs along retinal vessels and should therefore be part of the BV region, the boundaries of manually labeled BV and VB masks may not perfectly match

because of annotation variation and the subtle appearance of vessel edges. To obtain a reliable VB-positive candidate region, an intersection operation is applied between the BV mask and the VB annotation mask. This ensures that only pixels labeled as both vessel pixels and VB pixels are used as positive patch centers.

The positive candidate mask M_{pos} is defined as

$$M_{pos} = M_{BV} \cap M_{VB}$$

where M_{BV} denotes the BV ground-truth mask and M_{VB} denotes the VB annotation mask. This intersection-based strategy removes VB-labeled pixels that fall outside the manually labeled vessel region and improves anatomical consistency during patch extraction.

Negative candidate regions are generated from vessel pixels that do not overlap with the VB-positive region:

$$M_{neg} = M_{BV} \setminus M_{pos}$$

However, subtracting the VB-positive region from the BV mask may create small disconnected vessel fragments around VB areas. These small segments can introduce ambiguity when extracting negative patches near VB regions. Therefore, connected-component analysis is applied to the negative candidate mask using region properties, and the main connected BV region is retained for negative patch sampling. This step helps reduce interference from small fragmented regions near VB annotations and ensures that negative patches are sampled from stable non-VB vessel regions.

This localization procedure ensures that both positive and negative patches are anatomically meaningful, vessel-centered, and suitable for training a CNN classifier to distinguish VB from normal vessel structures.

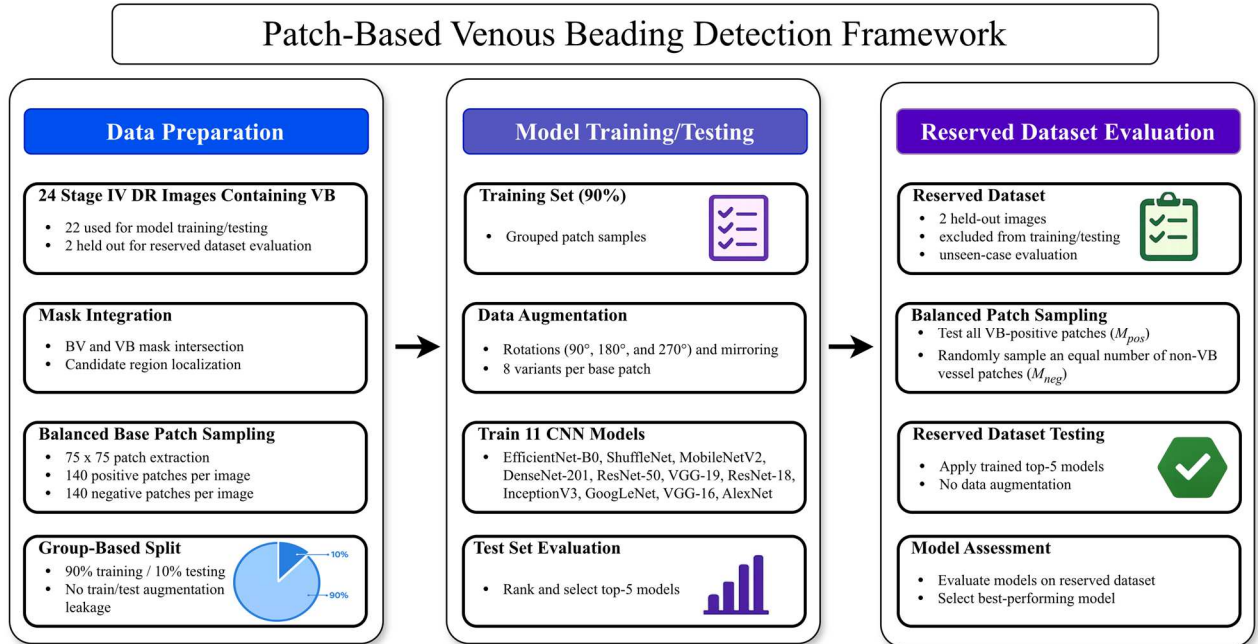


Fig. 2. Overview of the proposed patch-based venous beading detection framework. The workflow includes data preparation, model training/testing on the main patch dataset, and reserved dataset evaluation. Balanced patch sampling is used in both model development and reserved evaluation to reduce class imbalance, while the reserved dataset is kept separate to provide a limited held-out assessment on unseen cases.

D. Patch Extraction and Dataset Construction

After VB-positive and non-VB vessel candidate regions are identified, patch-level samples are extracted from the original RGB fundus images. Each patch is extracted as a 75×75 pixel window centered on a selected vessel pixel. For positive samples, patch centers are selected from M_{pos} , while negative samples are selected from the refined non-VB vessel region M_{neg} .

To avoid incomplete samples, boundary checking is applied during extraction. Candidate pixels located too close to the image border are excluded if a full 75×75 patch cannot be extracted. This ensures that all patches have the same spatial size and are suitable for CNN input after resizing to the required network input dimension.

For the main patch dataset, up to 140 positive and 140 negative base patches are extracted from each selected image. Each base patch is expanded into eight variants, including the original patch, three rotations (90° , 180° , and 270°), and mirrored versions, producing up to 1,120 positive and 1,120 negative augmented patches per image.

For the main patch dataset, the train/test split is performed at the base-patch group level. All augmented variants derived from the same base patch remain in the same partition, preventing augmented versions of the same patch from appearing in both training and testing sets.

E. Model Training and Evaluation

Eleven CNN architectures are evaluated for patch-level VB classification: EfficientNet-B0 [9], ShuffleNet [10], VGG-19 and VGG-16 [11], MobileNetV2 [12], DenseNet-201 [13], ResNet-50 and ResNet-18 [14], GoogLeNet [15], Inception-v3 [16], and AlexNet [17]. All models use the Adam optimizer [18] with a learning rate of 1×10^{-4} , and each 75×75 patch is resized to match the input size required by the corresponding network.

The evaluation is conducted in two stages. First, all eleven models are trained and tested on the main patch dataset to compare baseline performance. In this stage, training and testing sub-images are sampled from the 22 non-reserved images without overlap. Second, the top five models from the main patch dataset evaluation are tested on the reserved dataset to provide an additional limited held-out assessment on unseen images. For reserved evaluation, each selected model is first applied to all M_{pos} -based VB-positive sub-images. For each image, an equal number of M_{neg} -based non-VB vessel sub-images is then randomly sampled for testing, producing a balanced positive/negative evaluation set and reducing metric bias from the larger number of non-VB vessel regions. Performance is measured using accuracy, precision, recall, F1-score, sensitivity, and specificity.

III. EXPERIMENTAL RESULTS

This section reports the experimental results from the main patch dataset and the reserved dataset. Model performance is measured using accuracy, precision, recall, F1-score, sensitivity, and specificity. Accuracy measures overall correctness; precision measures the reliability of VB-positive predictions;

recall, equivalent to sensitivity, measures detection of actual VB-positive patches; specificity measures non-VB recognition; and F1-score balances precision and recall.

A. Main Patch Dataset Evaluation

Table I reports the training and test accuracy of all eleven CNN architectures on the main patch dataset. All models are trained using the same experimental configuration, including 75×75 vessel-centered patches, deterministic augmentation, and the Adam optimizer [18] with a learning rate of 1×10^{-4} .

TABLE I. ACCURACY COMPARISON ON MAIN PATCH DATASET

Model	Training Accuracy (%)	Test Accuracy (%)
EfficientNet-B0	100.00	97.42
ShuffleNet	100.00	96.67
MobileNetV2	100.00	96.35
DenseNet-201	100.00	96.31
ResNet-50	100.00	96.06
VGG-19	96.88	95.88
ResNet-18	100.00	95.43
Inception-v3	100.00	95.09
GoogLeNet	100.00	94.42
VGG-16	96.88	94.16
AlexNet	96.88	94.05

As shown in Table I, EfficientNet-B0 achieves the highest test accuracy at 97.42%, followed by ShuffleNet at 96.67%, MobileNetV2 at 96.35%, DenseNet-201 at 96.31%, and ResNet-50 at 96.06%. These results show that modern CNN architectures can effectively learn patch-level venous beading features from localized vessel regions.

B. Reserved Dataset Evaluation

To provide an additional limited held-out assessment, only the top five models from the main patch dataset evaluation are tested on the reserved dataset. This stage examines whether models that perform well on the main patch dataset can maintain performance on unseen images. Table II reports the reserved dataset results for the five selected models.

TABLE II. MODEL PERFORMANCE ON RESERVED DATASET

Metric	Models				
	A	B	C	D	E
Accuracy (%)	76.34	71.39	70.87	68.84	67.59
Precision (%)	83.59	86.55	82.41	86.09	83.14
Recall (%)	65.55	50.66	53.08	44.94	44.14
F1-score (%)	73.48	63.91	64.57	59.05	57.66
Sensitivity (%)	65.55	50.66	53.08	44.94	44.14
Specificity (%)	87.13	92.13	88.67	92.74	91.05

Note: A: ResNet-50; B: MobileNetV2; C: EfficientNet-B0; D: DenseNet-201; E: ShuffleNet.

As shown in Table II, ResNet-50 achieves the strongest balanced performance in this limited reserved dataset evaluation, with the highest accuracy, recall, sensitivity, and F1-score. MobileNetV2 and DenseNet-201 achieve higher precision and specificity, indicating more conservative classification behavior with fewer false positives. However, their lower recall values indicate that they miss more VB-positive patches.

Overall, the reserved dataset evaluation shows that model selection should not rely only on test accuracy from the main patch dataset. While EfficientNet-B0 achieves the highest accuracy on the main patch dataset, ResNet-50 provides the strongest balanced performance in the limited reserved dataset evaluation.

C. Performance Analysis

The experimental results demonstrate that patch-based learning is effective for lesion-level venous beading detection. By focusing on vessel-centered local patches, CNN models capture fine-grained vascular morphology that may be overlooked in full-image classification approaches.

A key observation from the reserved dataset evaluation is that ResNet-50 achieves the most balanced performance across multiple metrics, despite not having the highest accuracy in the main patch dataset evaluation. This highlights the importance of using additional held-out assessment rather than relying only on test accuracy from the main dataset.

Across all models, specificity is higher than sensitivity. This trend suggests that the models exhibit conservative classification behavior on the limited reserved dataset, achieving strong non-VB recognition while maintaining moderate VB detection capability. While this behavior helps reduce false positives, further sensitivity improvement is needed for screening-oriented deployment. From a clinical perspective, this performance characteristic implies that the current framework is more suitable for validation or decision-support applications rather than primary screening, where high sensitivity is critical.

IV. DISCUSSION AND CONCLUSION

The results demonstrate that patch-based deep learning is effective for lesion-level venous beading detection in fundus images. By focusing on localized vessel morphology, the framework enables CNNs to capture fine-grained vascular patterns that may be overlooked by full-image classification.

Lightweight architectures achieve strong performance on the main patch dataset. EfficientNet-B0 obtains the highest accuracy, while ShuffleNet and MobileNetV2 also perform competitively. However, evaluation on the limited reserved dataset shows that peak test accuracy alone is not sufficient for model selection. ResNet-50 achieves the strongest balanced performance on reserved images, with 76.34% accuracy, 83.59% precision, and 73.48% F₁-score, suggesting better performance in this limited held-out assessment.

The reserved dataset results also show higher specificity than sensitivity, suggesting conservative classification behavior. This makes the current framework more suitable for validation or decision-support use than primary screening, where high sensitivity is critical. Because the reserved dataset contains only two held-out images, Table II should be interpreted as a limited held-out assessment rather than population-level generalization evidence.

In conclusion, this study presents a reproducible patch-based framework for automated venous beading detection using vessel masks, VB annotations, balanced patch extraction, and CNN classification. The results support localized lesion-level learning

as a promising strategy for interpretable diabetic retinopathy assessment. Future work will include sensitivity improvement, larger-scale validation, repeated sampling analysis, and comparison with segmentation- or attention-based lesion detection methods.

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